

Questions on NumPy Arrays

1. Create a NumPy array of integers from 10 to 50.
2. Create an array of 10 zeros and replace the fifth value with 1.
3. Create a 3x3 matrix with values from 1 to 9.
4. Create a NumPy array of 20 random integers between 1 and 100.
5. Create a 5x5 identity matrix.
6. Create a NumPy array filled with 7s of shape (4, 4).
7. Generate a 1D array of 20 linearly spaced elements between 0 and 1.
8. Create a 3x3 matrix with random float values between 0 and 1.
9. Reverse a given 1D NumPy array.
10. Reshape a 1D array of 12 elements into a 3x4 matrix.
11. Extract the second column from a 4x4 matrix.
12. Slice a 5x5 matrix to get the central 3x3 matrix.
13. Replace all odd numbers in an array with -1.
14. Get all even numbers from a NumPy array.
15. Flatten a 2D matrix into a 1D array.
16. Find the mean, median, and standard deviation of a NumPy array.
17. Create two arrays and perform element-wise addition, subtraction, multiplication, and division.
18. Multiply two matrices using NumPy dot product.
19. Find the minimum and maximum values in a NumPy array and their indices.
20. Normalize a NumPy array (rescale to 0–1).
21. Stack two 1D arrays vertically and horizontally.
22. Replace all elements greater than 10 in a NumPy array with 10.
23. Broadcast a 1D array to a 2D array of shape (3, 5).
24. Split a 1D array into 4 equal parts.
25. Sort a NumPy array by column or row.
26. Create a checkerboard pattern (8x8) using 0s and 1s.

27. Create a border of 1s around a 0 matrix (e.g., 5x5).
28. Find the common values between two arrays.
29. Remove duplicate values from a NumPy array.
30. Replace all NaN values in a NumPy array with the mean of the array .